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DIMENSION REDUCTION ON MEASURES OF IMPULSIVITY

A thesis presented to the Faculty of

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By: Christina T. Schulz

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Abstract

The main focus of this study was twofold: the first purpose being the assessment of convergent validity on six different measures of impulsivity, and the second being a comparison of dimension reduction techniques for quantitative data. Results from a Principal Components Analysis (PCA) were compared with results from a novel dimension technique known as Local Linear Embedding (LLE). LLE is an analysis of dimension reduction for nonlinear, high dimensional data. By computing neighborhood preserving embeddings, LLE aims to map newly constructed coordinates into a global coordinate structure of a lower dimension. Past research using LLE has solely been conducted on visual and auditory-oriented data. Thus, this is a novel approach, applying LLE to cognitive measures rather than visual or auditory data. This paper acts as a secondary analysis of data collected by a prior Masters student (Robles, 2016). The measures of impulsivity included in this study were: The Barratt Impulsiveness Scale (BIS), the Iowa Gambling Task (IGT), the UPPS-P Impulsive Behavior Scale, the Cued Go/No-Go Task, the Stroop Color-Word Interference Test, and the Delay and Probability Discounting Task (DPDT) – all tests performed in a Latin-square determined order by each of 151 students. Findings not only indicate that convergent validity was absent for certain measures of impulsivity, but that with future programming, LLE may be a suitable method of dimension reduction for quantitative data.

Keywords: impulsivity, impulsiveness, dimension reduction, local linear embedding, principal components analysis

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Imagine a fluffy, white marshmallow sitting on the table in front of you. Now imagine someone telling you that if you wait until their return to eat the marshmallow, you can have not one, but two delicious treats in your possession. Could you control yourself for double the reward? Believe it or not, this test has been demonstrated as a predictor for impulsivity. It is known as the Marshmallow Test and it has been administered to children for over 40 years. For children who do consume the marshmallow before their instructor's return, poorer school performance, and an increased likelihood of substance abuse have been observed later in life (Mischel, 2014). One may wonder how a simple experiment could measure impulsivity with such precision. Another notion may be whether or not this test is actually measuring impulsivity.

The term “impulsivity” has been one of the most frequently applied and researched cognitive constructs (DeYoung, 2010), but it is not clear that all researchers are describing the same behaviors. According to Oas (1985), impulsivity may be described as quick and thoughtless behavior that is considered socially inappropriate. Others have described impulsivity as “a predisposition toward rapid, unplanned reactions to internal or external stimuli without regard to the negative consequences of these reactions to the impulsive individuals or to others” (Moeller, Barratt, Dougherty, Schmitz, & Swann, 2001, p. 1784). The Oxford English Dictionary in 2015 defined impulsivity as the act of behaving without reflection or forethought. Other behavioral traits typically associated with impulsivity include aggression, emotionally-charged action, and cognitive instability (Malloy-Diniz, Fuentes, Leite, Correa, & Bechara, 2007). Due to these various definitions, it may be suggested that the concept of “impulsivity” is actually an umbrella term, used to describe behaviors that may not necessarily co-occur.

These characteristic “behaviors” of impulsivity have typically been associated with specific psychological diagnoses. For instance, studies have indicated that impulsive behavior is

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characteristic of compulsive gambling (Alessi & Petry, 2003; Fuentes, Tavares, Artes, & Gorenstein, 2006). A key feature of impulsivity identified in gamblers is the preference of smaller, immediate rewards over larger, distant rewards. This preference for a smaller reward now instead of a larger reward later is also assessed in the Marshmallow Test (Mischel, 2011). Individuals identified as pathological gamblers have been described as restless, easily bored, and impulsive in addition to heightened sensation seekers (Brown, 1986). Impulsivity in individuals diagnosed with pathological gambling disorders has commonly been measured using psychometrically sound instruments, such as the Iowa Gambling Task (IGT) and the Delay Discounting Task. Other studies have found that greater levels of sensation seeking (another trait associated with impulsivity) and positive affect are predictive of pathological tendencies (Canale, Vieno, Griffiths, Rubaltelli & Santinello, 2015; Nower, Derevensky, & Gupta, 2004).

The dysregulation of a person's inhibitory system is implicated often as a key trait of impulsivity. Such dysregulation is a core symptom of Attention Deficit Hyperactivity Disorder (ADHD). Inhibitory dysregulation influences a range of behaviors that result in negative consequences. As mentioned above, many researchers believe that the disregard for an action's negative consequences is a major characteristic of impulsivity (Moeller et al., 2001). Lack of inhibition has also been associated with aggressive tendencies (Chester & DeWall, 2018; Mann & Yadav, 2016). Mann and Yadav (2016) believed there is an emotional component underlying the link between impulsivity and aggression where either arousal or intentions of revenge may lead to greater impulsive behavior. Neurotransmitter systems associated with impulsivity have also been found to be responsible for aggressive behaviors (Meyer & Lindenberg et al., 2006).

Substance use has also been connected to impulsivity in that impaired decision-making has been observed in these individuals. This impairment is believed to promote the addictive

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behavior despite the individual's health thus being compromised (Mitchell, Fields, D'Esposito & Boettiger, 2005). Prior studies investigating the role of impulsivity in substance use disorder have deployed very different instruments, which may not in fact be measuring the same construct. In one study, two separate instruments were used to investigate differences in impulsivity between comorbid cocaine users with personality disorders and individuals with cocaine use or personality disorder alone (Albein-Urios et al., 2013). The measures were the Stroop Color-Word Interference Test, a measure of cognitive inhibition of verbal habit, and the UPPS-P, a self-report scale that measures various dimensions of impulsivity. Although individuals with comorbidity demonstrated greater impulsivity on the UPPS-P, differences between the groups did not meet significance on the Stroop Test. That the tests did not yield parallel results suggests that they may not be different operationalizations of the same construct.

Another study utilized the BIS-11, a self-report scale similar to the UPPS-P, to investigate impulsivity in outpatients with disorders arising from cocaine abuse. Impulsivity was higher in individuals that suffered from cocaine-induced psychosis than in those that did not (Roncero, et al., 2010). Although similarities between the BIS-11 and UPPS-P have been found, the two differ in the dimensions that they measure. The BIS-11 assesses impulsivity based on the subscales *attentional impulsivity*, *motor impulsivity*, and *planning impulsivity* while the UPPS-P focuses on the subscales *emotion-based rash action*, *sensation seeking*, and *deficits in conscientiousness*.

Rogers, Moeller, Swann and Clark (2010) investigated levels of impulsivity between cocaine users and healthy controls using the Iowa Gambling Task (IGT). The IGT assesses decision-making and risky behavior by having participants select from advantageous and disadvantageous decks of cards. Results demonstrated that cocaine users showed heightened

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impulsivity based on their choices on the IGT task when compared to their healthy cohorts.

Another study investigated impulsivity among individuals with early- and late-onset alcohol abuse using the Delay Discounting Task. As previously mentioned, the Delay Discounting Task instructs participants to select either a smaller, immediate reward or a larger, delayed reward.

Participants in this study with early-onset alcohol abuse demonstrated higher discount rates when compared to late onset individuals. Higher discount rates have been associated with greater levels of impulsivity (Dom, D'haene, Hulstijn & Sabbe, 2006).

As indicated by these studies, diverse populations have demonstrated increased scores of impulsivity based on different assessments of impulsivity. Although findings are broadly consistent among these instruments, it is still not clear whether there is a common factor shared by these measures as they all appear to be measuring different behaviors. Because impulsivity is characterized by behaviors that may rarely co-occur, it would be reasonable to suggest that an instrument meant to assess “impulsivity” may not be appropriate for all populations. That is, impulsivity may be best assessed for an individual with ADHD using an instrument that may not be appropriate to use for someone with an addiction to alcohol because of the different “impulsive” behaviors associated with each. In order to investigate this unresolved question, it would be pertinent to analyze data collected from various measures of “impulsivity” using a model of dimension reduction.

The Current Study

Once considered unidimensional, impulsivity is a construct that research suggests may be an umbrella term that is entailed in domains such as decision-making, inhibition, and emotion to name a few. Designed to assess these specific traits, a number of instruments have been developed. However, the question remains as to what extent these different tasks can produce the same result.

In order to investigate this lingering query, a novel approach to dimension reduction is also explored, using Local Linear Embedding (LLE). Results from this analysis are compared to results from another dimension reduction technique (Principal Component Analysis, PCA) in order to confirm whether LLE may be an appropriate analysis for this type of data.

The goals of this study are twofold. They are to examine the relationship(s) between measures of impulsivity and to investigate the usage of Local Linear Embedding for quantitative data. Based on the literature, it would appear that impulsivity is made up of multiple components. This study is important for elucidating whether certain measures of impulsivity are appropriate to use for the assessment of certain traits. If these measures are found to not be related, then it would be reasonable to suggest that future studies should be more selective in their choosing of instruments, as one may be more appropriate over another when assessing specific populations. In addition, if Local Linear Embedding is found to be an appropriate technique for dimension reduction, future research may benefit from using this technique in that it does demonstrate an advantage over other methods in its preservation of local structure.

Methods

Participants and Procedure

The data used in this secondary analysis were collected on the campus of The City College of New York in Fall 2015 and Spring 2016 (Robles, 2016). A total of 151 introductory psychology students were recruited from the college's Department of Psychology subject pool. Sixteen participants were excluded from the analyses due to missing data. An initial screening assessment was conducted to determine eligibility. Recruitment criteria included English proficiency determined by a score of 15 or higher on the Cambridge English Proficiency Exam, and no history of motor deficits that could impede reaction time.

Although specific demographic information was not made available for the purposes of this study, the introductory psychology courses at the college, having approximately 800 students per semester, it would have been highly likely that the class demographics reflected that of the university. The gender of students was nearly split by males (49.7%) and females (50.3%). Approximately 77% of students were aged 24 years and younger. In regards to ethnicity, approximately 36% of students were reported Hispanic, 24.5% were Asian or Pacific Islander, 15.3% were Caucasian, 16.5% were African American. There were also a reported 23 Native American students (0.17%). Table 1 reflects the college population according to City Facts (2015).

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Table 1. City College Demographics

Students	N	%
Total Undergraduates	13,340	100
Gender		
Male	6,627	49.7
Female	6,713	50.3
Ethnicity		
Caucasian	2,044	15.3
African American	2,196	16.5
Hispanic	4,778	35.8
Native American	23	0.17
Asian or Pacific Islander	3,279	24.5
Age		
Under 20	3,994	29.9
20 to 22	4,550	34.1
23 to 24	1,693	12.7
25 to 29	1,754	13.1
30 to 44	1,072	8.1
45 or more	277	2.1

Participants were tested in a designated lab on the campus. Participants received course credit for their participation, and the Institutional Review Board of the university approved the

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study and all procedures. In order to avoid order effects, participants were assigned to one of six alternate task-orders. These task orders may be observed in Table 2. All six measures were administered to each participant on a computer using the Inquisit 4 software. The total time for each trial lasted approximately one hour and 15 minutes.

Table 2. Task Orders

BIS-11	UPPS-P	IGT	DPDT	Go/No-Go	Stroop
UPPS-P	IGT	DPDT	Go/No-Go	Stroop	BIS-11
IGT	DPDT	Go/No-Go	Stroop	BIS-11	UPPS-P
DPDT	Go/No-Go	Stroop	BIS-11	UPPS-P	IGT
Go/No-Go	Stroop	BIS-11	UPPS-P	IGT	DPDT
Stroop	BIS-11	UPPS-P	IGT	DPDT	Go/No-Go

Materials

The session consisted of 6 different assessments of impulsivity: the BIS-11, the UPPS-P, the Delay and Probability Discounting Task, the Stroop Color-Word Interference Test, the Cued Go/No-Go Task, and the Iowa Gambling Task.

Barratt Impulsiveness Scale (BIS-11). The Barratt Impulsiveness Scale (BIS; Barratt, Patton, & Stanford, 1975) is a self-report measure of impulsivity that was originally designed to accommodate the belief that impulsivity was a unidimensional construct. However, as time elapsed, adjustments were made to the scale and the original BIS was later revised to accommodate the “action-oriented” traits of impulsivity. This revised BIS-10 (Barratt, 1985) involved the following three subscales: *cognitive impulsiveness* which suggested rash decision, *motor impulsiveness* which suggested acting without thought, and *nonplanning impulsiveness*

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which suggested a lack of planning for the future. Later, after principal components analysis (PCA) was used to analyze BIS-10 scores on a sample of 412 college students, researchers concluded that items could be sorted into six separate 1st order factors, with each belonging to one of three revised subscales. This adaption of the BIS is the BIS-11 and is the version of the BIS used in this study (for the BIS-11, see Appendix A).

Thought to be the most frequently used assessment for impulsivity (Patton, Stanford, & Barratt, 1995), the BIS-11 is a 30-item questionnaire containing questions that relate to one of the three aforementioned revised subscales (also known as 2nd order factors) and takes fewer than five minutes to complete. The first subscale of the BIS-11 is concerned with *attentional impulsivity* (inability to sustain focus or concentration). Specific first-order factors belonging to this subscale include questions pertaining to *attention* (e.g. “I am a steady thinker” and “I concentrate easily”) and *cognitive instability* (e.g. “I change hobbies” and “I often have extraneous thoughts when thinking”). The second subscale is denominated as *motor impulsivity* (acting without thought) and contains questions pertaining to the 1st order factors of *motor* (e.g. “I do things without thinking” and “I make-up my mind quickly”) and *perseverance* (e.g. “I change jobs” and “I change residences”). The third and final subscale is associated with *nonplanning impulsivity* and includes the 1st order factors of *self-control* (e.g. “I plan tasks carefully” and “I say things without thinking”) and *cognitive complexity* (e.g. “I save regularly” and “I like puzzles”). Participants are asked to relate to these statements on a 4-point Likert scale with one being Rarely/Never and four being Almost Always/Always. Greater assessments of impulsivity on the BIS-11 are typically characterized by higher reported scores. Participants are also instructed not to dwell on each question but to answer “quickly and honestly.”

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In prior studies, internal consistency has been confirmed for the BIS-11, with Cronbach's alpha ranging from .79 to .83 (Patton et al., 1995). However, internal consistency was low for this study for items in both their respective 1st order and 2nd order scales. For the entire set of items Cronbach's alpha was .18, which demonstrates low internal consistency for the measure as a whole. Second order scales demonstrated low Cronbach alpha coefficients for this study: Attention (.42), Motor (-.00), and Nonplanning (.04). First order scales also demonstrated low Cronbach alpha coefficients for this study: Attention (.21), Cognitive Instability (.55), Motor (.00), Perseverance (-.25), Self-Control (.02) and Cognitive Complexity (.22). For this reason, each item was assessed in this study's dimension reduction analyses. This method is not unique as scoring for the BIS-11 has varied in recent publications. Some researchers (e.g.,) have chosen to measure impulsivity by calculating a total for the 30 items while others (e.g.,) have used first- and second- order scores to measure impulsivity.

UPPS-P Impulsive Behavior Scale. The Urgency, Premeditation (lack of), Perseverance (lack of), Sensation Seeking, Positive Urgency Impulsive Behavior Scale (UPPS-P; Lynam, Whiteside, Smith, & Cyders, 2006) is a 59-item self-report instrument that, like the BIS-11, is based on the belief that impulsivity is a multidimensional construct. The original UPPS scale initially measured impulsivity by assessing four personality factors (1st Order): Negative Urgency (frequent rash behavior under the pressure of negative emotions), Lack of Premeditation (acting without thought), Lack of Perseverance (difficulty sustaining focus), and Sensation Seeking (desire to seek out thrill-evoking experiences; Whitehead & Lynam, 2001). The UPPS was later revised to accommodate Positive Urgency (frequent rash behavior under extreme positive emotions) as the dimension was not well represented in the literature. For the UPPS-P, see Appendix B.

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These aforementioned first-order factors, again like the BIS-11, contain questions that coincide with three specific second-order factors. Both Negative Urgency (e.g. “When I am upset I often act without thinking”) and Positive Urgency (e.g. “I tend to lose control when I am in a great mood”) belong to the factor of Emotion Based Rash Action, Lack of Premeditation (e.g. “I usually make up my mind through careful reasoning”) and Lack of Perseverance (e.g. “I tend to give up easily”) fall into Deficits in Conscientiousness, and Sensation Seeking (e.g. “I generally seek new and exciting experiences and sensations”) falls onto its own 2nd Order Factor, also labeled “Sensation Seeking”. Participants are asked to relate to these statements on a 4-point Likert scale with one being Agree Strongly and four being Disagree Strongly. Similar to the BIS-11, greater scores on the UPPS-P suggests higher levels of impulsivity.

Internal consistency for the UPPS-P in this study was strong with a Cronbach alpha of .93 for the entire scale. Second order scales also demonstrated strong Cronbach alpha coefficients for this study: Emotion Based Rash Action (.93), Sensation Seeking (.79), Deficits in Conscientiousness (.84). All first order factors demonstrated strong coefficients as well: Negative Urgency (.87), Positive Urgency (.87), Lack of Premeditation (.80), and Lack of Perseverance (.79). In regards to scoring, prior studies have used different methods. Akin to the BIS-11, some studies used the total score, while others used either the first- or second- order scale scores. For this study, each item will be assessed in the dimension reduction analyses.

Stroop Color-Word Interference Test. The Stroop Color-Word Interference Test (Stroop, 1935) is a measure that assesses the inhibition of verbal habit. The task involves four separate conditions (C). In the first condition (C1), participants are asked to promptly name different color patches. In the second condition (C2), participants are asked to read the names of colors which are presented in black lettering. The third condition (C3) is again composed of

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color names, however they are presented in incongruent coloring (e.g. the word *Green* is written with red ink). In C3, participants are instructed to name the color the word is written in and ignore the word itself. The Stroop Effect occurs when participants incorrectly read the color name rather than the ink color. Lastly, condition four (C4) is composed of the same color swatches as C3, however, they are asked to alternate between reading the color name or the ink color itself. For each condition, participants are asked to correct any error they may commit. Scores for each individual condition are measured using each participant's completion time (and/or accuracy). A total inhibition score is calculated by subtracting the completion time on C1 from the completion time on C3 (Albein-Urios, Pilatti, Lozano, Martinez-Gonzalez & Verdejo-Garcia, 2013).

Because the Stroop Test has been used to examine verbal inhibition, it has been a common measure for impulsivity with a higher inhibition score associated with greater impulsivity. However, due to pressures placed on the participant, impulsivity may not be the construct measured by this instrument. Pressure may produce errors for the participant, and thus provide an incorrect assessment of impulsivity for the individual. Longer completion times may actually symbolize weaker levels of impulsivity, in that participants took their time to read the correct color, rather than hastily commit an error.

Cued Go No-Go Task. The Cued Go/No-Go Task (Fillmore, 2003) is an instrument designed to assess impulsive motor control. It is a behavioral task in which participants are cued with a go or no-go cue prior to an actual go or no-go target is displayed. Such cues provide information as to the probability of whether either the go or no-go target will appear in the imminent trial. After being presented with a fixation cross, a white rectangle (cue; horizontal or vertical) is shown to participants who have been instructed to press a computer key upon seeing a

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green box (go target) and to abstain from pressing the computer key upon seeing a blue box (no-go target). Five SOAs for cue times are randomized (100, 200, 300, 400, or 500ms). The cue-target relationship is altered so that there is a higher probability of correctly responding to a go or no-go target (valid cues), and a lower probability of incorrectly responding to a target (invalid cue). Invalid cues are typically associated with flawed response inhibition and slower response execution. The critical trials observed in this task are the invalid cue trials where a go cue is presented to the participant with a no-go target to follow. Weaker inhibitory control is measured in these trials by counting the number of failures to inhibit their responses that the participant accumulates. This task contains 250 trials and take approximately 15 minutes to complete.

Delay and Probability Discounting Task. The Delay and Probability Discounting Task (DPDT; Richards, Zhang, Mitchell, de Wit, 1999) is a behavioral measure typically used to assess impulsivity in those with gambling addictions. Impulsivity in the DPDT is operationally defined as the degree to which one opts for smaller-more immediate rewards over larger-delayed ones (Ainslie, 1975). For delay trials, participants are presented with a choice of an immediate amount of money or \$10 after a delay (i.e., “Would you rather have \$2 now or \$10 in 15 days?”). Similarly, in probability trials, participants are asked to choose from a certain amount of money or \$10 with a given probability (i.e., “Would you rather have \$5 at the end of this session or \$10 with a 50% chance?”).

A random adjusting-amount procedure (Richards et al., 1999) determines the immediate/certain amount for each trial. Discounting is measured for six delays (0, 2, 30, 180, and 365 days later) and five probabilities (25%, 50%, 75%, 90%, 100%). The random adjusting-amount procedure mentioned above results in the creation of an indifference point for each delay and probability interval. These indifference points represent the subjective value of the amount

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of impulsivity at a given trial. Prior studies have assessed these indifference points using either a hyperbolic or exponential function. For the purposes of the present study, the Area Under the Curve (AUC) for both delay and probability is used. The AUC has been a valid model in a number of recent studies (Myerson, Green, & Warusawitharana, 2001; Du, Green, & Myerson, 2002; Ohmura, Takahasi, & Kitamura, 2005). The areas under these discounting curves (AUC) are calculated by adding the resulting trapezoids. Greater AUC curves are indicative of lesser discounting and thus, lower impulsivity.

Iowa Gambling Task (IGT). The computerized Iowa Gambling Task (IGT; Bechara, 2007) is a decision-making task believed to measure the same aspect of the DPDT described above. In the IGT, participants are provided with an initial virtual \$2000 four decks of cards, and instructs them to select a card from any deck where they will either earn or lose money. The four decks differ in their risk/reward payoff, with decks A and B yielding greater rewards but with higher loss rates. The safer decks (decks C and D), however, have a lower overall loss rate, but produce smaller earnings. Decks A and B are considered risky because of their “disadvantageous” terms while decks C and D are believed to be the more optimal of choice. Participants are presented with 20-block trials over 100 card plays. For this study, impulsivity is assessed using the IGT by the total amount of money “won” by the participant, with the greater the amount earned, the lower the assumed level of impulsivity.

Although each of these measures is believed to measure impulsivity, based on these descriptions, it may be suggested that each of these instruments actually measure different components. For instance, the UPPS-P is a self-report scale that measures numerous facets of impulsivity such as emotion and planning while the Iowa Gambling Task is a decision-making

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task based on risk. How accurately impulsivity is measured between the two may vary because they focus on separate components.

Dimension Reduction Approaches and Terminology.

Dimension reduction is a process by which a large number of correlated variables are reduced into a smaller set of principal variables (Fodor, 2002). The most common technique of dimension reduction is a global method known as Principal Components Analysis (PCA). In terms of the method being global, PCA attempts to preserve geometry on all scales by mapping nearby points on a manifold to nearby points on a low dimensional representation. The method does the same in regards to faraway points. A local method, however, attempts to preserve geometry by simply mapping nearby points on a manifold with nearby points onto a lower dimensional space. Through PCA, a large number of correlated variables are combined to generate a smaller subset of components that are uncorrelated and retain as much of the variance in the data as possible. Specifically, PCA generates linear combinations of the variables that are calculated containing the majority of the variance using the highest eigenvectors of the data's covariance matrix (Silva & Tenenbaum, 2003).

Alternatively, Local Linear Embedding (LLE; Roweis & Saul, 2000) is an unsupervised learning algorithm for high dimensional, nonlinear dimension reduction. Typically used to analyze pattern recognition with images and speech, LLE calculates the neighborhood-preserving embeddings of high dimensional data and maps newly constructed coordinates into a global coordinate structure of a lower dimension. There are advantages to LLE compared to other dimension reduction models such as MDS and PCA. For instance, there are no global constraints for LLE as there are for these previous models as there is no longer a need to estimate

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distances among widely separated points with LLE. The advantage of LLE over PCA in the matter of local preservation can be seen in Figure 1.

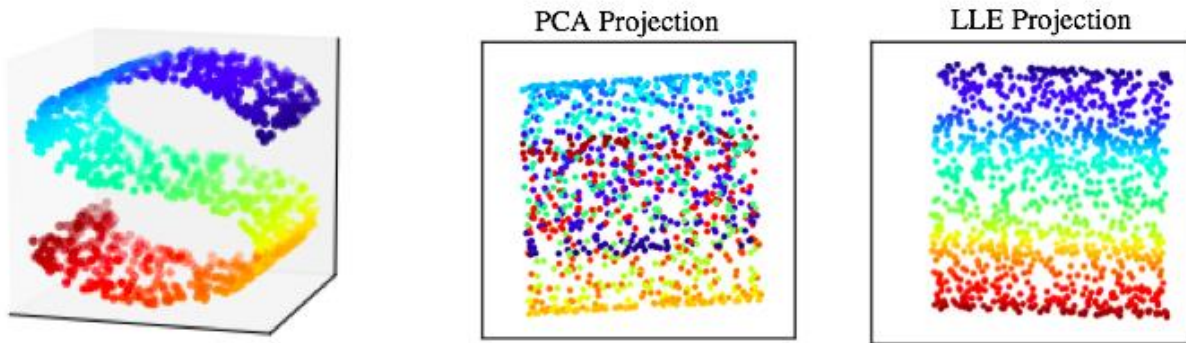


Figure 1. Two projections of S-curve data set using PCA (center) and LLE (far right) onto a two-dimensional manifold. A blending of colors can be seen in the PCA projection while color delineation is apparent in the LLE projection. This delineates an advantage of LLE as local data points are preserved, rather than the global approach of the PCA.

There are a series of steps to employ with the LLE algorithm (see Figure 2; Roweis & Saul, 2000). Step one involves calculating the distance from each point in the data set (x_i) to every other point (x_j). By doing this, the smallest (but optimal) value for K (neighbors) may be found. K may also be found using an epsilon-ball method which would contain all points within a restricted radius or by using KD trees. The second step involves calculating reconstruction weights (w). In order to calculate the weights a third matrix (Z) must be created containing each neighbor of x_j . x_i must then be subtracted from each column in Z . The local covariance is then computed ($C = Z^T * Z$) as is a linear system ($C * w = 1$) for w ($W_{ij} = 0$ if j is not a neighbor of i). The remaining values in the i^{th} row of W are then set equal to sum of w . The third step involves computing the embedded coordinates (Y) using the weights by creating a sparse matrix ($M = (I - W)^T * (I - W)$; $M = \text{transpose of Matrix } M$). The lower $d+1$ eigenvectors of M (which correspond

to the $d-1$ smallest eigenvectors) are calculated and the q^{th} row of Y (output matrix) is set to the $q+1$ smallest eigenvector.

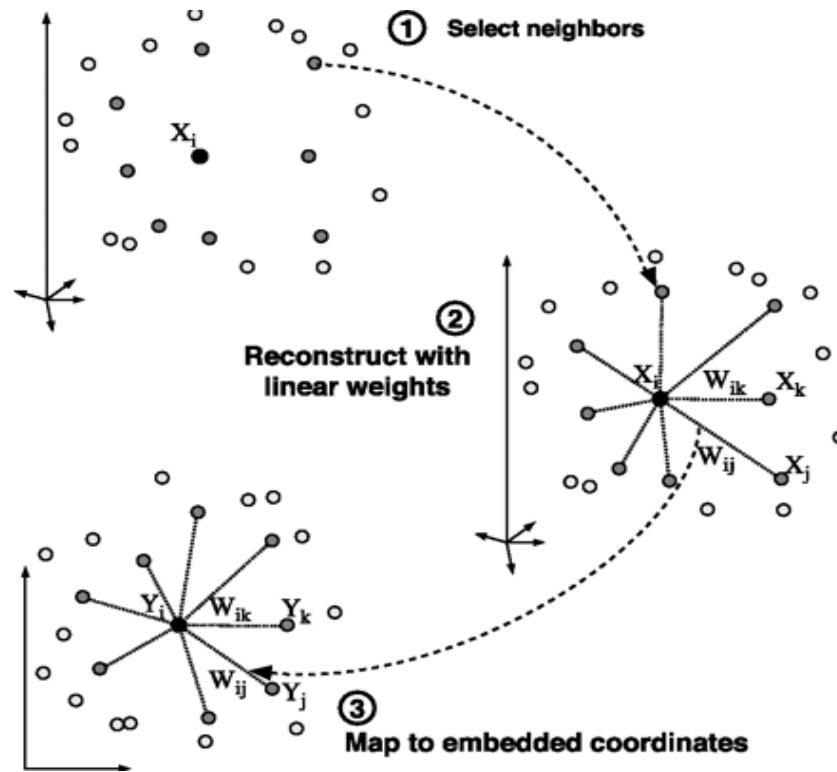


Figure 2. The three steps of Local Linear Embedding. (1) Select Neighbors for each data point after computing optimal value for k , (2) Reconstruct Linear Weights using chosen neighbors, (3) Map the embedded coordinates onto a lower dimensional manifold (here it is 2-dimensional; Roweis & Saul, 2000)).

Results

Inter-item reliability was assessed for both the BIS-11 and UPPS-P instruments. Internal consistency was low for items on the BIS-11 for items in both their respective 1st order and 2nd order scales. For the entire set of items, Cronbach's alpha was .18, which demonstrates low internal consistency for the measure as a whole. Second order scales demonstrated low Cronbach alpha coefficients for this study: Attention (.42), Motor (-.00), and Nonplanning (.04). First order

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scales also demonstrated low Cronbach alpha coefficients for this study: Attention (.21), Cognitive Instability (.55), Motor (.00), Perseverance (-.25), Self-Control (.02) and Cognitive Complexity (.22). Internal consistency for the UPPS-P in this study was strong with a Cronbach alpha of .93 for the entire scale. Second order scales also demonstrated strong Cronbach alpha coefficients for this study: Emotion Based Rash Action (.93), Sensation Seeking (.79), Deficits in Conscientiousness (.84). All first order factors demonstrated strong coefficients as well: Negative Urgency (.87), Positive Urgency (.87), Lack of Premeditation (.80), and Lack of Perseverance (.79).

I. Descriptive Statistics & Correlations

A total of 54.5% of the participants earned over the average amount gained on the Iowa Gambling Task ($M = 1816.80$, $SD = 845.5$). Greater earnings indicate more conservative betting on the task, thus more than half of the sample appeared to demonstrate lesser impulsivity based on this task. For the Go-No Go task, 11.2% of the sample committed more errors on the no-go trials than average ($M = .05$, $SD = .15$). This indicates that a small percentage of the sample demonstrated heightened impulsivity based on this task as more frequent errors indicated greater impulsivity. Based on the Stroop Interference score, 44.8% of the sample took longer than average to complete incongruent trials ($M = 222.3$, $SD = 244.9$). This demonstrates that, based on this task, almost half of the sample demonstrated heightened impulsivity. Over 50% of the sample demonstrated greater discounting on both the delay AUC score ($M = .64$, $SD = .89$) and probability AUC score ($M = .62$, $SD = .79$). The average scores for the UPPS-P subscales were reported: *Negative Urgency* ($M = 34.4$, $SD = 7.3$), *Positive Urgency* ($M = 43.8$, $SD = 6.9$), *Sensation Seeking* ($M = 27.9$, $SD = 6.2$), *Lack of Premeditation* ($M = 35.8$, $SD = 4.6$), and *Lack of Perseverance* ($M = 31.2$, $SD = 4.4$).

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Prior to LLE analysis to provide evidence of nonlinearity, correlations among variables were calculated. A Pearson product-moment correlation coefficient was computed to assess the relationship between the variables involved in this study. Certain items on the BIS-11 were found to be significantly correlated with items found within their own measure as well as with subscales on the UPPS-P. One reason for this may be due to the overlapping questions contained within both measures. Correlation matrices are displayed in Tables 3 & 4.

Table 3. *Correlations between Measures of Impulsivity.*

Measure	1	2	3	4	5	6	7	8	9	10
1. Iowa Gambling Task	—									
2. Go/No-Go Task	-.056	—								
3. Stroop Interference	-.043	.163	—							
4. DPDT Delay	.129	.021	.191*	—						
5. DPDT Probability	.126	.016	.205*	.997**	—					
6. UPPS-P Negative Urgency	.008	.016	.078	-.152	-.153	—				
7. UPPS-P Positive Urgency	.047	-.044	.028	-.196*	-.201*	.807**	—			
8. UPPS-P Sensation Seeking	.011	.039	-.034	-.022	-.022	.181*	.359**	—		
9. UPPS-P Lack of Premeditation	.021	-.077	-.017	-.184*	-.192*	.503**	.568**	.327**	—	
10. UPPS-P Lack of Perseverance	-.012	.107	.158	-.203*	-.198*	.479**	.369**	-.027	.408**	—
<i>M</i>	1816.8	.05	222.3	.64	.62	34.4	43.8	27.9	35.8	31.2
<i>SD</i>	845.5	.15	244.9	.89	.79	7.3	6.9	6.2	4.6	4.4

Note: * = significant at .05 level

** = significant at .01 level

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Table 4. *Correlations between Measures of Impulsivity ctd.*

Measures	IGT	Go/No-Go	Stroop	DPDT Delay	DPDT Probability	UPPS-P NU	UPPS-P PU	UPPS-P SS	UPPS-P LPM	UPPS-P LPS
BIS-11										
I plan tasks carefully.	.126	-.002	.129	.943**	.940**	-.258**	-.328**	-.073	-.327**	-.329**
I do things without thinking.	.132	.032	.188*	.996**	.993**	-.127	-.171*	-.071	-.153	-.193*
I make-up my mind quickly.	.017	.110	-.111	-.272**	-.278**	.112	.112	.069	.244**	.103
I am happy-go-lucky.	-.193*	-.080	-.126	-.220*	-.221*	-.104	-.029	.077	.039	-.153
I don't "pay attention."	-.057	-.072	-.015	-.312**	-.312**	.368**	.332**	.228**	.370**	.417**
I have "racing" thoughts.	-.088	.085	-.068	-.163	-.175*	.171*	.229**	.128	-.032	.038
I plan trips well ahead of time.	.011	.005	-.067	-.140	-.149	-.118	-.110	.093	-.122	-.226**
I am self-controlled.	-.020	-.032	-.248**	-.269**	-.265**	-.506**	-.401**	-.077	-.315**	-.324**
I concentrate easily.	.046	-.003	.029	-.121	-.113	-.245**	-.167	.005	-.168	-.385**
I save regularly.	.256**	-.070	-.068	-.192*	-.186*	-.046	-.026	-.027	-.096	-.105
I "squirm" at plays or lectures.	-.066	-.113	-.049	-.399**	-.395**	.214*	.256**	.006	.048	.139
I am a careful thinker.	-.025	-.070	-.067	-.042	-.042	-.271**	-.181*	-.116	-.426**	-.302**
I plan for job security.	.023	.047	-.059	-.221*	-.217*	-.122	-.051	.032	-.150	-.155
I say things without thinking.	.005	-.031	-.054	.082	.085	.387**	.422**	.258**	.385**	.138
I like to think about complex problems.	.183*	.251**	-.144	-.115	-.121	.014	.055	.149	.040	-.055
I change jobs.	-.036	.099	-.115	-.098	-.095	.096	.048	.133	.048	.238**
I act "on impulse."	-.049	.013	-.008	.084	.079	.378**	.373**	.260**	.365**	.250**
I get easily bored when solving thought problems.	.126	-.038	.050	-.220*	-.213*	.301**	.258**	-.018	.087	.334**
I act on spur of the moment.	-.080	.071	-.044	-.007	-.010	.403**	.452**	.293**	.326**	.156
I am a steady thinker.	-.143	.002	.059	.125	.129	-.351**	-.271**	.002	-.411**	-.358**
I change residences.	.037	.052	-.073	.053	.037	.060	.086	.143	.133	.014
I buy things on impulse.	-.171*	-.029	-.055	.104	.101	.272**	.299**	.073	.163	.093
I can only think about one thing at a time.	.017	-.071	.080	.080	.084	.049	.088	-.157	-.126	.031
I change hobbies.	.124	.069	-.167	-.039	-.043	.309**	.287**	.138	.275**	.229*
I spend or charge more than I earn.	-.107	-.108	-.055	.072	.068	.309**	.278**	.105	.130	.237**
I often have extraneous thoughts when thinking.	.069	.133	.007	.128	.119	.319**	.363**	.180*	.199*	.192*
I am more interested in the present than the future.	-.111	-.200*	.090	-.186*	-.180*	.291**	.266**	.052	.129	.175*
I am restless at the theater or lectures.	-.024	-.035	-.045	-.038	-.040	.381**	.360**	-.048	.159	.296**
I like puzzles.	.004	-.025	-.034	-.114	-.113	-.0677	-.055	.189*	-.016	-.143
I am future oriented.	.016	.055	-.090	-.088	-.088	-.241**	-.240	.145	-.201*	-.296**

Note: NU = Negative Urgency; PU = Positive Urgency; SS = Sensation Seeking; LPM = Lack of Premeditation; LPS = Lack of Perseverance

* = significant at .05 level

** = significant at .01 level

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As Table 4 summarizes: four BIS-11 items were found to be significantly correlated with the Iowa Gambling Task (“I am happy-go-lucky” (-.193), “I save regularly” (.256), “I like to think about complex problems” (.251), and “I buy things on impulse” (-.171)). The Stroop Test was significantly correlated with the DPDT probability score ($r = .205, p < .05$), and the DPDT delay score ($r = .191, p < .05$). In addition, there was a significant correlation between the Stroop Test and one item found on the BIS-11 (“I am self-controlled” (-.248)). No significant relationships were found between the Stroop Test and any other measure. Two items on the BIS-11 (“I like to think about complex problems” (.251), “I am more interested in the present than the future” (-.200)) were found to be significantly correlated with the Go/No-Go task. The Go/No-Go task was not found to be correlated with any other measure. As predicted, the scores for the DPDT delay and probability were significantly correlated with one another ($r = .997, p < .01$).

The DPDT delay score was also found to be significantly correlated with the UPPS-P subscales of *Positive Urgency* (-.196), *Lack of Premeditation* (-.184), and *Lack of Perseverance* (-.203). Significant relationships were also found between the DPDT delay score and 11 items on the BIS-11. The DPDT probability was also significantly correlated with the UPPS-P subscales of *Positive Urgency* (-.201), *Lack of Premeditation* (-.192) and *Lack of Perseverance* (-.198) in addition to 11 items on the BIS-11.

II. Principal Component Analysis

First, the factorability of the 40 variables were examined. The Kaiser-Meyer-Olkin measure of sampling adequacy was .72, above the commonly recommended value of .6, and Bartlett’s test of Sphericity was significant ($\chi^2 (780) = 3308.60, p < .01$). The communalities were all above .3, confirming that each item shared a common variance with at least one other

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item. Due to these conditions being met, factor analysis appeared to be an appropriate method to continue with.

Principal component analysis was used because a purpose of this study was to compare measures of impulsivity as well as to compare methods of dimension reduction. Initial eigen values indicated that the data could be reduced to 13 components (C). The solutions for the 24 factors were each examined using a varimax rotation of the factor loading matrix. The 13 factor solution, which explained approximately 70% of the variance, was selected due to the Kaiser criterion (Kaiser, 1960) which recommends retaining factors with an eigen value greater or equal to one.

Out of the initial 40 variables, 39 were retained because they each met a minimum criterion of having a primary factor loading of .4 or above. The one item from the BIS-11 that did not load was “I plan trips well ahead of time.” Cross loadings above .4 were found for two items on the BIS-11 (“I “squirm” at plays or lectures,” “I like to think about complex problems,” “I buy things on impulse”) and for the Sensation Seeking subscale on the UPPS-P. To mediate this, the greater loading for each score was favored. Out of the 13 components, there were four single factor loadings. For component 9, BIS-11 item “I like puzzles” was the sole loading and accounted for 3.3% of the variance. The second single factor loading component 11 (C11), contained the Stroop Test (.794) which accounted for 3% of the variance. The third single factor loading was C12 which contained the Go/No-Go measure (-.848) and represented 2.7% of the variance. C13 is the final single loading component and contained BIS-11 item “I change residences.” This loading represented 2.5% of the variance.

C1 explained 16% of the variance and consists of the DPDT Delay score (.976), the DPDT Probability Score (.974), and two BIS-11 items (“I plan tasks carefully” (.961) and “I do

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things without thinking” (.972). C2 explained 12% of the variance and consisted of BIS-11 items (“I say things without thinking,” “I act “on impulse”,” “I act on spur of the moment,” and “I am self-controlled”) as well as four UPPS-P subscales (*Negative Urgency*, *Positive Urgency*, *Sensation Seeking*, and *Lack of Premeditation*). C3 accounts for 7% of the variance and contains four BIS-11 items (“I am happy-go-lucky,” “I concentrate easily,” “I am a careful thinker,” and “I am a steady thinker”) as well as the UPPS-P subscale *Lack of Perseverance* (-.468). C4 accounted for 5% and contained loadings solely of BIS-11 items (“I have racing thoughts,” “I “squirm” at plays or lectures,” “I get easily bored when solving thought problems,” and “I often have extraneous thoughts when thinking”). C5 accounted for 4.7% of the variance and also contained strictly BIS-11 items (“I plan for job security,” “I like to think about complex problems,” “I am more interested in the present than the future,” and “I am future oriented”). The sixth component contained loadings pertaining to the Iowa Gambling Task (-.499) as well as three BIS-11 items (“I save regularly,” “I buy things on impulse,” and “I spend or charge more than I earn”) and accounts for 4.3% of the variance. C7 contained two BIS-11 items (“I can only think about one thing at a time” and “I am restless at the theater or lectures”) and accounts for 3.7% of the variance. C8 contained three BIS-11 loadings (“I like to think about complex problems,” “I change hobbies,” and “I make-up my mind quickly”) and accounts for 3.4% of the variance. The tenth component had loadings pertaining to BIS-11 items “I change jobs” and “I don’t pay attention”. The two account for 3.1% of the variance. The factor loading matrix for this solution may be observed in Appendix C.

III. Local Linear Embedding (LLE)

Due to findings of nonlinearity for the majority of the variables included, it was determined that Local Linear Embedding (LLE; Roweis & Saul, 2000) is an appropriate analysis

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to continue with in this study. The analysis was performed using the LLE code written by Vanderplas and Connolly (2009). Three values for the optimal number of neighbors were calculated for the data ($K = 8, 9$, and 12) with $K = 12$ being the preferred choice (using R studio). This decision was based on trial and error using swiss roll templates as seen in Figure 3. When applying $K = 8$ to both a standard Swiss roll (Figure 3a) and S curve (Figure 3b) algorithm using the same number of samples as this study's data set, a poor embedding was discovered and does not reflect the topographical structure (see Figure 1). Similar findings were discovered when using $K = 9$. The topographical mapping was reflected when using $K = 12$, and demonstrated a solid embedding.

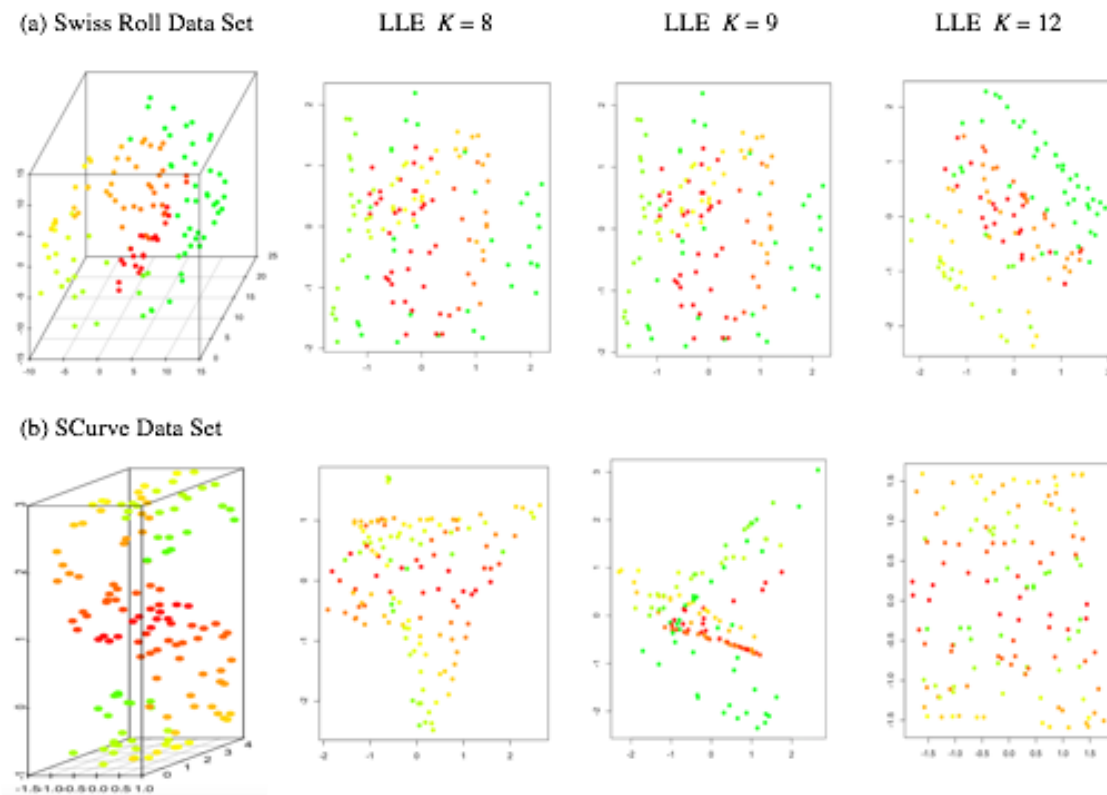


Figure 3. A value of $K = 12$ was chosen for the optimal number of neighbors to use for the data set based on the plotting of standard Swiss Roll and S-Curve algorithms using $N = 134$. Preservation of local embeddings was best kept using $K = 12$ for both sets.

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After selecting a value for K , analysis continued as a neighborhood was built for each point (x_i) in the data followed by a calculation of weights for the linear approximation of the data for each neighborhood. By applying the weights, low dimensional coordinates were found. Three-dimensional plotting for the PCA analysis was also conducted in order to compare the two methods. They can be seen in Figure 5. For the PCA analysis, each of the first three components is represented by a specific axis (Component 1, x-axis; Component 2, y-axis; Component 3, z-axis). For the LLE analysis, each of the axes represents three eigenvectors that the analysis was instructed to condense the data into.

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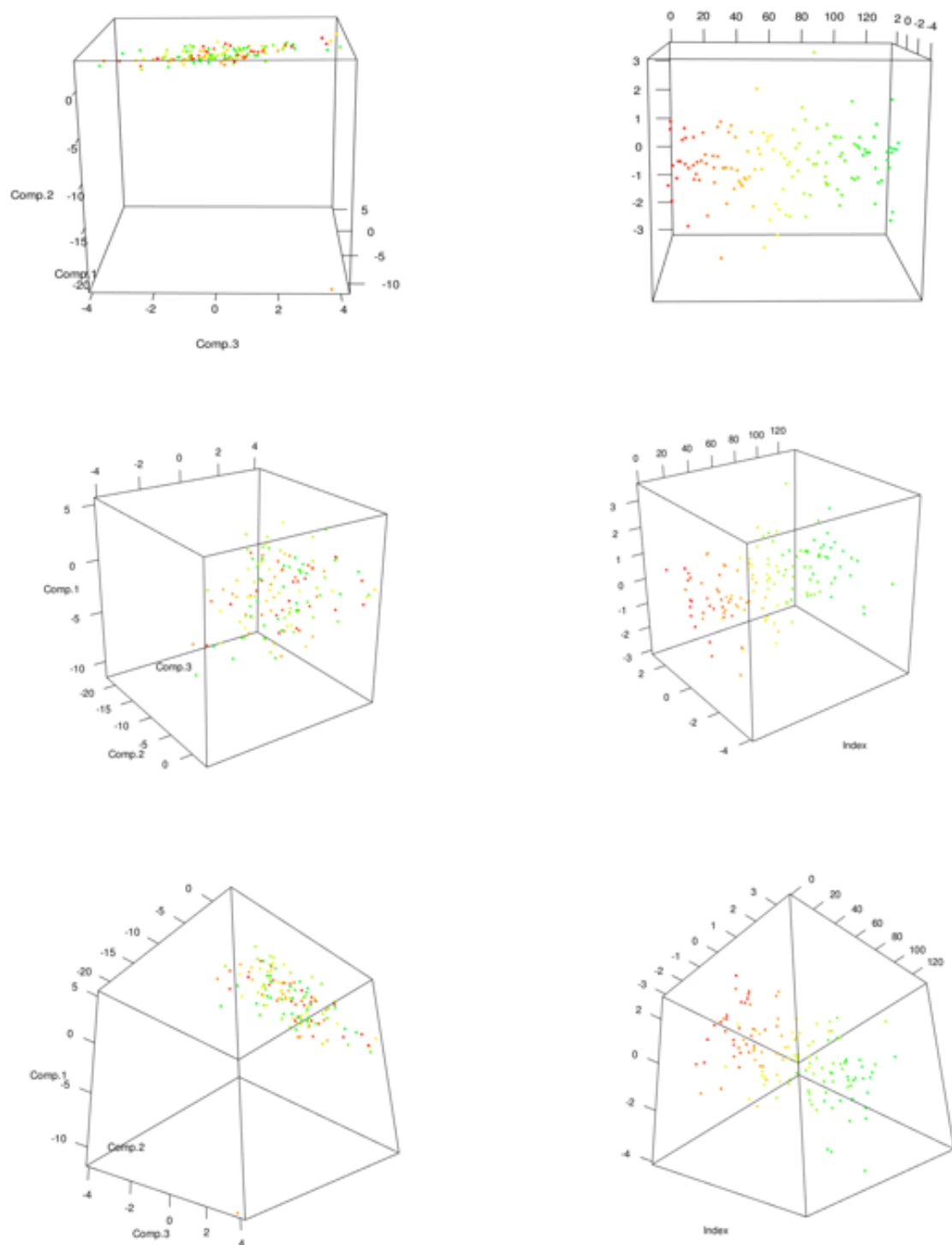


Figure 5. Three dimensional projections from an LLE analysis (right) juxtaposed against three dimensional projections from a PCA analysis (left) of the same data.

Discussion

There were two research questions that this study aimed to answer: whether measures of impulsivity actually assess the same construct and whether local linear embedding (LLE) is an appropriate technique of dimension reduction for this type of data.

Impulsivity Conclusions

To review: the word “impulsivity” appears to be an umbrella term that has been used to described diverse cognitive and behavioral properties alike. In this study, data from six separate supposed measures of impulsivity was collected in an effort to determine whether similar assessments would result. Based on preliminary analyses, internal consistency was low for the Barratt Impulsiveness Scale and high for the UPPS-P for the entire scale as well as 1st- and 2nd order subscales. This finding may indicate that-for this sample size-the BIS-11 may be too brief as it contains 30 items while the UPPS-P contains 59 (Tavakol & Dennick, 2011). This low consistency may also reflect the possibility that some items that are believed to be related, actually are not. Although internal consistency was low, some items on the BIS-11 were found to be strongly related to items believed to belong to 1st- and 2nd-order subscales that were not their own. For instance, certain items believed to belong to the *Motor* subscale, were found to be significantly correlated with items on the *Non-planning* subscale such as the items “I plan tasks carefully” and “I do things without thinking” which had a correlation coefficient close to 1. However, it would be important to note that psychometrics of the BIS-11 have been sound in prior research where the sample size was much greater than the sample taken in this study (Stanford et al., 2009).

Apart from items on the BIS-11, all of the UPPS-P subscales were significantly correlated with one another, except for the *Sensation Seeking* and *Lack of Premeditation* scales.

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Although these items do not correspond to the same subscales, it may have been predicted that the two scales would result in a positive relationship as *Sensation Seeking* coincides with seeking out thrilling experiences and *Lack of Premeditation* is associated with acting without forethought.

Both the Iowa Gambling Task (IGT) and the Cued Go/No-Go were not significantly related to any other measure apart from items on the BIS-11. However, it is important to note that the IGT was related to BIS-11 items “I save regularly” and “I buy things on impulse.” This suggest that the IGT is accurate in the assessment of financial impulsivity whereby greater earning on the IGT which represents conservative choices, is positively correlated with an individual’s saving habits and negatively correlated with impulsive buying. If this is the explanation, “financial impulsivity” constitutes a different construct with different behavioral ramifications from other types of impulsivity.

Both the delay and probability scores from the Delay and Probability Discounting Task (DPDT) were strongly correlated with one another, which was expected based on prior findings (Green, Myerson, & Vandervelt, 2014; Ohmura, Takahashi, Kitamura, & Wehr, 2006). Both scores were also significantly related to the Stroop Color Word Interference Test as well as three UPPS-P scales (*Positive Urgency*, *Lack of Premeditation*, and *Lack of Perseverance*). This finding is interesting because the DPDT is an untimed decision making task while the Stroop is a timed verbal inhibition task. It would be pertinent to consider, however, that the correlation, although significant, is very weak.

In regards to dimension reduction, 13 components were extracted from the Principal Components Analysis. Only one variable did not load onto any component: the BIS-11 item “I plan trips well ahead of time.” Variables that loaded into the first component and accounted for

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the majority of the variance included both the DPDT delay and probability scores as well as two BIS-11 items (“I plan tasks carefully” and “I do things without thinking”). This loading may suggest that the DPDT and these two items can be used to describe financial or investment impulsivity.

The second component retained the greatest number of variables and contained the BIS-11 items of “I say things without thinking,” “I act “on impulse”,” “I act on spur of the moment,” and “I am self-controlled” as well as four UPPS-P subscales (*Negative Urgency, Positive Urgency, Sensation Seeking, and Lack of Premeditation*). The BIS-11 items suggest acting without forethought; however, because of the UPPS-P scales being retained, it might be difficult to identify an underlying component. These loadings may also suggest that these items on the BIS-11, which are identified under the Motor and Self-Control subscales, are related to these UPPS-P subscales. However, Motor and Self-Control are not specified in the design of the UPPS-P.

The fifth subscale of the UPPS-P (*Lack of Perseverance*) was retained in the third component along with four BIS-11 items (“I am happy-go-lucky,” “I concentrate easily,” “I am a careful thinker,” and “I am a steady thinker”). The wording of these items suggests the underlying component here to be attention. However, the items belong to four separate subscales on the BIS-11. Based on these findings and the matter of low consistency for this BIS-11 in this study, we suggest that certain items on this measure may have been misidentified when assigning them to a subscale.

The forth component only contained four BIS-11 items (“I have racing thoughts,” “I “squirm” at plays or lectures,” “I get easily bored when solving thought problems,” and “I often have extraneous thoughts when thinking”). These items correspond to the 2nd order subscales of

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Attentional and *Nonplanning* and may be appropriately labeled as *Cognitive Complexity* as it is titled in the BIS-11.

The fifth component retained four BIS-11 items (“I plan for job security,” “I like to think about complex problems,” “I am more interested in the present than the future,” and “I am future oriented”). These loadings suggest an underlying component of future planning. Interestingly, they do not all belong to the same subscales on the BIS-11. This again indicates that items on the BIS-11 may not have been appropriately designated to subscales.

The Iowa Gambling Task (IGT) was retained in the sixth component, as well as three BIS-11 items (“I save regularly,” “I buy things on impulse,” and “I spend or charge more than I earn”). Again financial impulsivity is a common factor for these variables. Financial impulsivity appears to be a recurrent factor in a number of these variables. It is odd, though, for items found in component one and component six to have not loaded onto the same component due to their common traits.

Two BIS-11 items loaded onto component seven (“I can only think about one thing at a time” and “I am restless at the theater or lectures”). One may connect these items in terms of stimulation. Individuals who identified strongly with the former statement may find stimulation from a show or a lecture to be over-bearing. Although the two items belong to two separate subscales on the BIS-11, one may infer from this loading that the two actually do group together. Interestingly, these items did not load with other items that appeared more relatable. For instance, “I am restless at the theater or lectures” did not load onto the same component as “I squirm” at plays or lectures.” The two items, although significantly correlated, were also weakly related.

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The eighth component contained three BIS-11 loadings (“I like to think about complex problems,” “I change hobbies,” and “I make-up my mind quickly”). Again, these items do not appear to be associated with one another. With all three items belonging to separate 1st factor subscales on the BIS-11, it is even more difficult to identify an underlying component.

The tenth component had loadings pertaining to BIS-11 items “I change jobs” and “I don’t pay attention”. The item “I change jobs” may appear vague when relating it to impulsivity. That is, it does not appear to be sufficiently specific when identifying it as an assessment of impulsivity. A more appropriate term may be “I change jobs often” or to even eliminate it from the test.

There were four single factor loadings in the PCA analysis. “I like puzzles (C9)” from the BIS-11 was one, and similar to the aforementioned “I change jobs,” its affiliation with impulsivity does not seem evident. This may indicate why it was a single factor loading, as it does not relate to any other variable in the data set. A second single-factor loading contained the Stroop Test (C11). This may be due to its underlying component of measuring verbal inhibition. The only other measure included in this analysis that measured inhibition was the Go/No-Go measure which also belonged to a single-factor loading (C12). The final single loading was component thirteen which contained the BIS-11 item “I change residences.” Like “I change jobs” and “I like puzzles,” this item’s inclusion in the BIS-11 is vague and may not be appropriate for assessing impulsivity.

These findings indicate that there do not appear to be underlying common components within each and every measure analyzed in this study. Based on these findings, it may be reasonable to suggest that the term “impulsivity” is a construct that should be carefully considered when selecting an instrument for assessment. That is, if one were to assess financial

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impulsivity or risk, it may be more pertinent to administer the Iowa Gambling Task rather than the Stroop Color Word Interference Task, which would be more suitable for verbal or cognitive inhibitory analyses.

Dimension Reduction Conclusions

In regards to local linear embedding, it would appear that there is still both research to conduct and programming to create before this method can be applied to quantitative data. After conducting principal components analysis, local linear embedding was pursued in an effort to reach similar conclusions. As discussed, LLE is a method of dimension reduction typically used to map visual and aural stimuli. Because there are no global constraints, it is believed that LLE takes advantage over previous methods of dimension reduction.

When results from both the LLE and PCA analysis are compared, it would seem that PCA is a more legible or understandable interpretation of the data. PCA is capable of clearly indicating the variables that load onto each component, while LLE has not yet reached this step. However, because there was a clear separation among measures within the three-dimensional plots (see Figure 5) it would appear that local structure was preserved in the LLE analysis. This is based on the delineation of colors demonstrated with LLE while colors are blended in the PCA analysis projection.

Based on this mapping, it may be reasonable to suggest that LLE as a potential method for dimension reduction in quantitative data. However, due to the lack of specificity, the measures that relate or that are even redundant cannot yet be identified through this analysis. Although a lack of clarity is evident for this analysis, these results do demonstrate that with future programming, local linear embedding may be a suitable technique of dimension reduction for quantitative data.

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These findings not only expand the existing literature on local linear embedding, but also provide an introduction into the use of LLE for quantitative data. Prior studies have applied LLE to either visual or aural data; however, this study investigated and confirmed that quantitative data may also benefit from the advantages that LLE has over other methods of dimension reduction. With further research and advanced programming, LLE may be a more appropriate and advantageous method of dimension reduction for quantitative data.

Limitations

There are limitations to this study, however. Because internal consistency was low for the BIS-11, a greater sample size may or may not have affected the loadings. Future research may be imperative for the reorganization of BIS-11 items if findings are replicated with a greater sample size. Other research should investigate which specific measures of impulsivity are more appropriate for certain populations. It is important to note that the current study aimed to identify whether instruments of impulsivity were assessing the same construct. This study did not aim to identify- for example- whether the Stroop or a self-report scale such as the BIS-11 or the UPPS-P would be a more suitable measure of impulsivity for individuals diagnosed with ADHD. Future research should aim to identify and distinguish these measures and their suitability for specific populations. Further, in regards to Local Linear Embedding, a more advanced programming is still necessary for better interpretation of the data. That is, current programming lacks specificity for LLE where detailed interpretation for components that one would see in PCA, is not yet available for LLE.

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Appendix A - Barratt Impulsiveness Scale (BIS-11)

DIRECTIONS: People differ in the ways they act and think in different situations. This is a test to measure some of the ways in which you act and think. Read each statement and put an X on the appropriate circle on the right side of this page. Do not spend too much time on any statement. Answer quickly and honestly.				
①	②	③	④	
Rarely/Never	Occasionally	Often	Almost Always/Always	
1 I plan tasks carefully.	①	②	③	④
2 I do things without thinking.	①	②	③	④
3 I make-up my mind quickly.	①	②	③	④
4 I am happy-go-lucky.	①	②	③	④
5 I don't "pay attention."	①	②	③	④
6 I have "racing" thoughts.	①	②	③	④
7 I plan trips well ahead of time.	①	②	③	④
8 I am self controlled.	①	②	③	④
9 I concentrate easily.	①	②	③	④
10 I save regularly.	①	②	③	④
11 I "squirm" at plays or lectures.	①	②	③	④
12 I am a careful thinker.	①	②	③	④
13 I plan for job security.	①	②	③	④
14 I say things without thinking.	①	②	③	④
15 I like to think about complex problems.	①	②	③	④
16 I change jobs.	①	②	③	④
17 I act "on impulse."	①	②	③	④
18 I get easily bored when solving thought problems.	①	②	③	④
19 I act on the spur of the moment.	①	②	③	④
20 I am a steady thinker.	①	②	③	④
21 I change residences.	①	②	③	④
22 I buy things on impulse.	①	②	③	④
23 I can only think about one thing at a time.	①	②	③	④
24 I change hobbies.	①	②	③	④
25 I spend or charge more than I earn.	①	②	③	④
26 I often have extraneous thoughts when thinking.	①	②	③	④
27 I am more interested in the present than the future.	①	②	③	④
28 I am restless at the theater or lectures.	①	②	③	④
29 I like puzzles.	①	②	③	④
30 I am future oriented.	①	②	③	④

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Appendix B - UPPS-P Impulsive Behavior Scale

UPPS-P

Below are a number of statements that describe ways in which people act and think. For each statement, please indicate how much you agree or disagree with the statement. If you **Agree Strongly** circle **1**, if you **Agree Somewhat** circle **2**, if you **Disagree somewhat** circle **3**, and if you **Disagree Strongly** circle **4**. Be sure to indicate your agreement or disagreement for every statement below. Also, there are questions on the following pages.

	Agree Strongly	Agree Some	Disagree Some	Disagree Strongly
1. I have a reserved and cautious attitude toward life.	1	2	3	4
2. I have trouble controlling my impulses.	1	2	3	4
3. I generally seek new and exciting experiences and sensations.	1	2	3	4
4. I generally like to see things through to the end.	1	2	3	4
5. When I am very happy, I can't seem to stop myself from doing things that can have bad consequences.	1	2	3	4
6. My thinking is usually careful and purposeful.	1	2	3	4
7. I have trouble resisting my cravings (for food, cigarettes, etc.).	1	2	3	4
8. I'll try anything once.	1	2	3	4
9. I tend to give up easily.	1	2	3	4
10. When I am in great mood, I tend to get into situations that could cause me problems.	1	2	3	4
11. I am not one of those people who blurt out things without thinking.	1	2	3	4
12. I often get involved in things I later wish I could get out of.	1	2	3	4
13. I like sports and games in which you have to choose your next move very quickly.	1	2	3	4
14. Unfinished tasks really bother me.	1	2	3	4
15. When I am very happy, I tend to do things that may cause problems in my life.	1	2	3	4
16. I like to stop and think things over before I do them.	1	2	3	4
17. When I feel bad, I will often do things I later regret in order to make myself feel better now.	1	2	3	4
18. I would enjoy water skiing.	1	2	3	4
19. Once I get going on something I hate to stop.	1	2	3	4
20. I tend to lose control when I am in a great mood.	1	2	3	4
21. I don't like to start a project until I know exactly how to proceed.	1	2	3	4

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	Agree Strongly	Agree Some	Disagree Some	Disagree Strongly
22. Sometimes when I feel bad, I can't seem to stop what I am doing even though it is making me feel worse.	1	2	3	4
23. I quite enjoy taking risks.	1	2	3	4
24. I concentrate easily.	1	2	3	4
25. When I am really ecstatic, I tend to get out of control.	1	2	3	4
26. I would enjoy parachute jumping.	1	2	3	4
27. I finish what I start.	1	2	3	4
28. I tend to value and follow a rational, "sensible" approach to things.	1	2	3	4
29. When I am upset I often act without thinking.	1	2	3	4
30. Others would say I make bad choices when I am extremely happy about something.	1	2	3	4
31. I welcome new and exciting experiences and sensations, even if they are a little frightening and unconventional.	1	2	3	4
32. I am able to pace myself so as to get things done on time.	1	2	3	4
33. I usually make up my mind through careful reasoning.	1	2	3	4
34. When I feel rejected, I will often say things that I later regret.	1	2	3	4
35. Others are shocked or worried about the things I do when I am feeling very excited.	1	2	3	4
36. I would like to learn to fly an airplane.	1	2	3	4
37. I am a person who always gets the job done.	1	2	3	4
38. I am a cautious person.	1	2	3	4
39. It is hard for me to resist acting on my feelings.	1	2	3	4
40. When I get really happy about something, I tend to do things that can have bad consequences.	1	2	3	4
41. I sometimes like doing things that are a bit frightening.	1	2	3	4
42. I almost always finish projects that I start.	1	2	3	4
43. Before I get into a new situation I like to find out what to expect from it.	1	2	3	4
44. I often make matters worse because I act without thinking when I am upset.	1	2	3	4
45. When overjoyed, I feel like I can't stop myself from going overboard.	1	2	3	4

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	Agree Strongly	Agree Some	Disagree Some	Disagree Strongly
46. I would enjoy the sensation of skiing very fast down a high mountain slope.	1	2	3	4
47. Sometimes there are so many little things to be done that I just ignore them all.	1	2	3	4
48. I usually think carefully before doing anything.	1	2	3	4
49. When I am really excited, I tend not to think of the consequences of my actions.	1	2	3	4
50. In the heat of an argument, I will often say things that I later regret.	1	2	3	4
51. I would like to go scuba diving.	1	2	3	4
52. I tend to act without thinking when I am really excited.	1	2	3	4
53. I always keep my feelings under control.	1	2	3	4
54. When I am really happy, I often find myself in situations that I normally wouldn't be comfortable with.	1	2	3	4
55. Before making up my mind, I consider all the advantages and disadvantages.	1	2	3	4
56. I would enjoy fast driving.	1	2	3	4
57. When I am very happy, I feel like it is ok to give in to cravings or overindulge.	1	2	3	4
58. Sometimes I do impulsive things that I later regret.	1	2	3	4
59. I am surprised at the things I do while in a great mood.	1	2	3	4

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Appendix C

Table 5. Factor loadings and communalities based on a principal components analysis with varimax rotation for 39 measures of Impulsivity (N = 134).

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13
BIS-11													
I plan tasks carefully.	.96												
I do things without thinking.	.97												
I make-up my mind quickly.								.44					
I am happy-go-lucky.			.62										
I don't "pay attention."										.50			
I have "racing" thoughts.				.73									
I am self-controlled.		-.57											
I concentrate easily.			.68										
I save regularly.						-.78							
I "squirm" at plays or lectures.				.50									
I am a careful thinker.			.56										
I plan for job security.					.66								
I say things without thinking.		.71											
I like to think about complex problems.				.42									
I change jobs.										.85			
I act "on impulse."		.71											
I get easily bored when solving thought problems.				.43									
I act on spur of the moment.		.76											
I am a steady thinker.			.72										
I change residences.													.812
I buy things on impulse.						.50							
I can only think about one thing at a time.							.83						
I change hobbies.								.69					
I spend or charge more than I earn.						.77							
I often have extraneous thoughts when thinking.				.67									
I am more interested in the present than the future.					-.61								
I am restless at the theater or lectures.							.60						
I like puzzles.									.79				
I am future oriented.					.81								
IGT						-.50							
Go/No-Go												-.85	
Stroop Interference											.79		
UPPS-P NU		.58											
UPPS-P PU		.65											
UPPS-P SS		.46											
UPPS-P LPS		.59											
UPPS-P LPM			-.47										
DPDT Delay	.98												
DPDT Probability	.97												

Note: Factor loadings <.4 are suppressed.

BIS = Barratt Impulsiveness Scale; UPPS-P NU = UPPS-P Negative Urgency; UPPS-P PU = UPPS-P Positive Urgency; UPPS-P SS = UPPS-P Sensation Seeking; UPPS-P LPS = UPPS-P Lack of Perseverance; UPPS-P LPM = UPPS-P Lack of